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ORIGINAL ARTICLE

Evaluation and correction of the TRMM 3B43V7 and GPM 3IMERGM satellite precipitation products by use of ground-based data over Xinjiang, China

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Abstract
Satellite retrieval of precipitation is a great challenge in arid regions where light precipitation prevails. To improve the accuracy of satellite precipitation products in Xinjiang, which is the driest region in China, this study first evaluated the performance of two widely used monthly satellite products from April 2014 to August 2017. The first was the Tropical Rainfall Measuring Mission (TRMM) 3B43 version 7 (hereafter 3B43V7), and the second was the Integrated Multi-Satellite Retrievals for the Global Precipitation Measurement (GPM) mission (IMERG) (hereafter 3IMERGM). The 3IMERGM product was then corrected by a stepwise regression model using topographic variables derived from digital elevation model (DEM). A comparison between satellite estimates and in situ measurements indicates that both 3B43V7 and 3IMERGM overestimate precipitation overall, but that 3IMERGM performs better than 3B43V7. The relative biases (RB) of 3B43V7 and 3IMERGM were found to be 10.24 and 7.76%, respectively. The correlation coefficient (CC) between 3IMERGM and the gauges is 0.68, which is higher than the CC observed (i.e., 0.62) between 3B43V7 and the gauges. To comprehensively evaluate the performance of the corrected model, a tenfold cross-validation method was used. The results showed that the corrected 3IMERGM (C-3IMERGM) performed much better than 3IMERGM. Specifically, CC was increased from 0.68 to 0.73, and RB is decreased from 7.76 to − 1.65%. Furthermore, C-3IMERGM achieves a better precipitation distribution than the uncorrected satellite product and even than scarce gauge measurements. The actual spatial pattern of precipitation represented that the precipitation bands in the Kunlun Mountains located in southern Xinjiang were captured by C-3IMERGM, but missed by the other products.

Keywords TRMM · GPM · Stepwise regression · Precipitation correction · Xinjiang

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Introduction

Precipitation is a vital component in the water and energy cycle and plays an important role in many research fields such as hydrology, meteorology, and climatology (Wu and Chen 2012; Tang et al. 2016c). However, accurate precipitation estimation remains a great challenge because of sparse gauge networks and the large spatial variability of precipitation, especially over mountainous regions, oceans, and polar areas. Moreover, gauges have difficulty in representing areal rainfall over a large region due to the inherent limitation of point measurements (Groisman and Legates 1994; Sinclair et al. 1997). Interpolation of rain gauge data inevitably introduces more uncertainty. Previous studies indicated that earth observation satellites are the only feasible approach to acquire integrated precipitation estimates on a global scale (Villarini and Krajewski 2008; Hou et al. 2014). In remote

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areas with few gauges and weather radars, satellite remote-sensing data are particularly valuable for supporting precipitation-relevant research and applications (Tang et al. 2016b).

In recent years, many satellite precipitation products with various spatiotemporal resolutions have been developed and released to the public. Examples of such products include Precipitation Estimation from Remotely Sensed Information using Artificial Neural Networks (PERSIANN) (Sorooshian et al. 2000), Climate Prediction Center (CPC) the MORPHing technique (CMORPH) (Joyce et al. 2004), TRMM Multi-satellite Precipitation Analysis (TMPA) (Huffman et al. 2007), Global Satellite Mapping of Precipitation (GSMaP; Kubota et al. 2007), and Integrated Multi-Satellite Retrievals for the Global Precipitation Measurement (GPM) mission (IMERG) (Huffman et al. 2017b). The GPM satellite is the successor of the TRMM satellite. A core satellite and a number of other satellites form the GPM constellation to offer next-generation global rain and snow products (Hou et al. 2014; Yong et al. 2015). Compared with TRMM, GPM incorporates three critical improvements: (1) greater coverage of climate zones because the orbital inclination angle of the GPM satellite is increased from 35° in the TRMM satellite to 65°, (2) greater sensitivity of dual-frequency precipitation radar (DPR) to light precipitation and snowfall, and (3) liquid and solid precipitation detection based on "high-frequency" passive microwave channels (165.5 and 183.3 GHz) (Huffman et al. 2017a). IMERG is the Level-3 product of the GPM mission (Huffman et al. 2017b) with $0.1° \times 0.1°$ spatial and half-hourly temporal resolutions. IMERG algorithms integrate the strong points of previous multi-satellite precipitation retrieval algorithms, including TMPA, CMORPH, and PERSIANN (Huffman et al. 2017a).

However, the accuracy of satellite precipitation products is generally lower than that of ground-based observations because of the indirect nature of observations, sampling error, retrieval algorithm uncertainty, and other factors (Ma et al. 2014; Yang and Luo 2014a). In arid regions where precipitation is characterized by low intensity and is difficult for satellite sensors to capture, satellite products can perform much worse than in humid regions (Tang et al. 2016b; Huang et al. 2016). Therefore, assessment and correction of satellite products are necessary, particularly for arid regions (Yin et al. 2004, 2008; Ji and Chen 2012; Yang and Luo 2014a, b; Ouallouche and Ameur 2014; Golian et al. 2015).

Xinjiang is the largest province in China and is characterized by complex topography and dry climate (Li et al. 2011a), both of which affect the quality of satellite precipitation estimates (Yang and Luo 2014b). Tang et al. (2016b) indicated that the correlation between IMERG products and ground observations is much lower in Xinjiang than in the eastern and southern parts of China. Both TRMM and GPM products fail to reproduce the diurnal cycle in Xinjiang because of the effect of dry climate. In addition, Guo et al. (2016) demonstrated that IMERG performs comparatively poorly in winter and overestimates light precipitation in northwestern China. Therefore, correction of satellite precipitation estimates is needed in Xinjiang.

It is well known that topography has a notable effect on precipitation (Smith 1979; Roe et al. 2002; Guan et al. 2005; Crochet 2009; Shige et al. 2013, 2014; Yamamoto and Shige 2015). Incoming moisture is forced to rise by mountains and hills, leading to more rainfall on the windward side and less rainfall on the leeward side (Barros and Lettenmaier 1992, 1994; Michaud et al. 1995). Various studies have investigated how to correct and improve existing satellite precipitation estimates according to regional terrain characteristics. Yin et al. (2004) found that merging terrain and geographic position factors significantly improved the accuracy of satellite precipitation estimates over the Tibetan Plateau (TP). The method has also been proven useful in correcting TRMM data over the TP (Yin et al. 2008). By incorporating a stepwise regression model and a back-propagation (BP) neural network including topographic and location factors over the arid area in northwestern China, the quality of TRMM 3B42 and 3B43 estimates was significantly improved (Yang and Luo 2014a). Xinjiang has rugged topography with three large mountain ranges, the Altay, Tianshan, and Kunlun Mountains (Fig. 1). The spatial distribution of precipitation is strongly dependent on topography, with less precipitation in low-lying basins and more precipitation along the windward slopes of mountains.

Therefore, this study has three objectives. The first is to evaluate the quality of TRMM 3B43V7 and GPM 3IMERGM in Xinjiang using the national meteorological stations as a reference. The second is to correct the latest 3IMERGM using a regression model integrating terrain characteristics and to compare the performance of 3IMERGM before and after correction. The third is to investigate the actual spatial distribution of precipitation over Xinjiang based on the C-3IMERGM dataset. This study should be valuable for understanding the uncertainty of satellite precipitation products and improving their quality in regions with arid climate and complex topography.

Materials and methods

Study area

The study area is Xinjiang, which is a typical arid and semiarid region located in northwestern China. Its spatial coverage is between 73°E–96°E and 34°N–49°N. Its area is 1,600,000 km^2, or about one-sixth of the land area in China (Fig. 1). The terrain is very complex, with three mountain ranges (the Altay, Tianshan, and Kunlun Mountains) and two basins (the Junggar and Tarim Basins). The elevation ranges

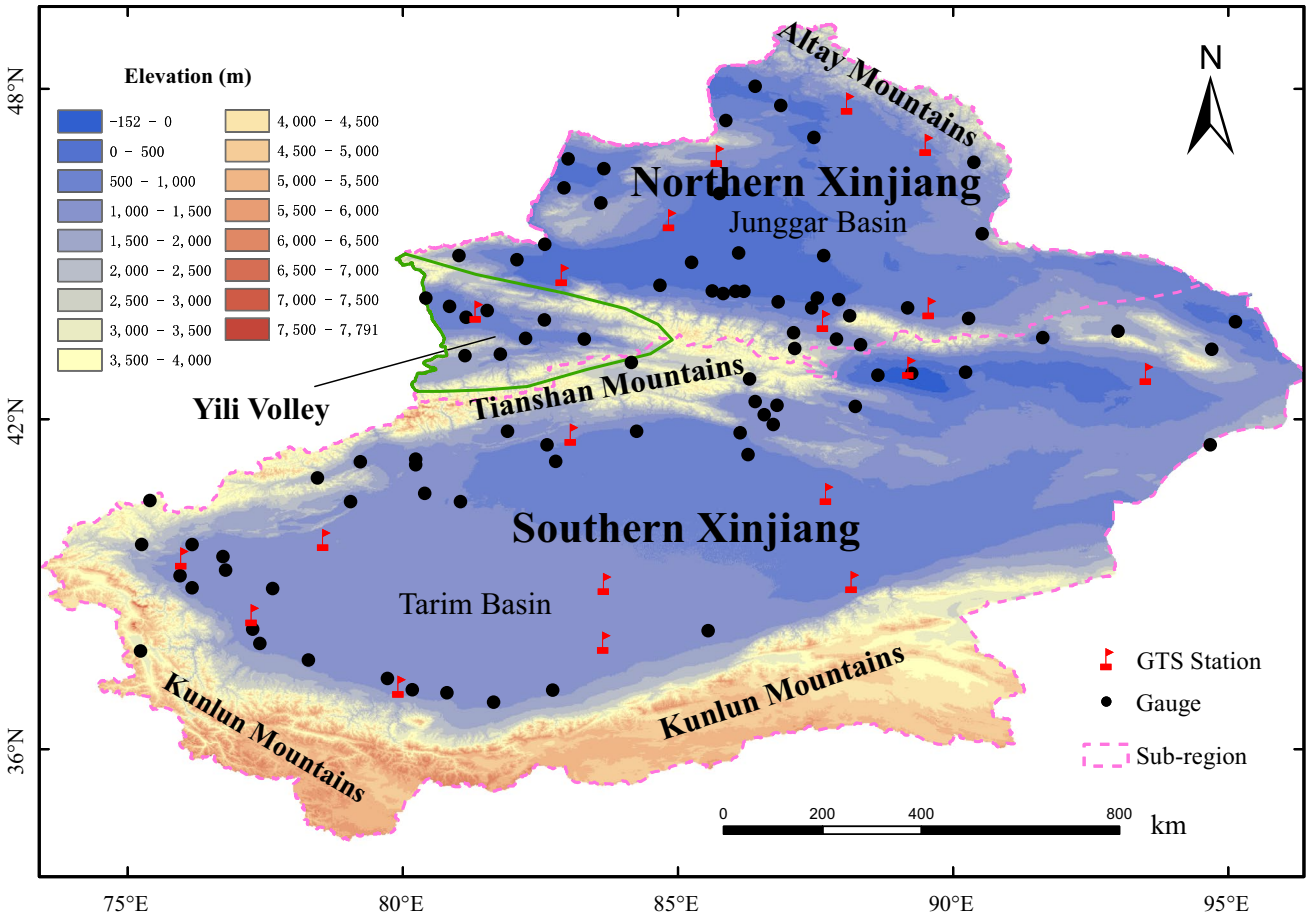


Fig. 1 Map of Xinjiang Province showing the DEM, the locations of the main mountains and basins, and the spatial distribution of rain gauges

from − 169 to 8611 m (mean sea level) MSL. The three mountain ranges from north to south are the Altay, Tianshan, and Kunlun Mountains, and their maximum elevations are 4373, 7455, and 8611 m, respectively (Zhao et al. 2011). There are only about 100 national meteorological stations in Xinjiang. The lowest altitude among them is − 47.4 m MSL, whereas the highest is 3507.2 m MSL. The altitudes of most gauge stations are less than 1500 m. Because of the sparse and nonuniform layout of the stations, the representation of precipitation is very deficient. Therefore, it is difficult to conduct studies on climate change and hydrological processes in Xinjiang.

Xinjiang is separated by the Tianshan Mountains in the middle of the region into two large areas: northern Xinjiang (NX) and southern Xinjiang (SX). There are two vast deserts within Xinjiang: the Taklimakan Desert and the Gurbantunggut Desert, located in the southern and northern part of Xinjiang, respectively. The former is located in the center of the Tarim Basin in SX, and the latter is located in the center of the Junggar Basin in NX. Being in the hinterland of Eurasia, far from the sea, Xinjiang has a typical temperate continental climate. Annual precipitation is only 150 mm, and the number of rainy days in 1 year is 55 on average. The spatial precipitation distribution is uneven, with annual precipitation of 210 mm in NX and less than 100 mm in SX. On the windward slopes of the Tianshan Mountains, annual precipitation can reach 500–700 mm (Zhang et al. 2012; Xu et al. 2016; Tan and Shao 2016).

Data sources

Satellite retrievals

This study used the newest Version 7 post-real-time 3B43 (3B43V7) data, which have 0.25° × 0.25° and monthly resolutions and spatial coverage between 50°N and 50°S. The 3B43V7 product provides the best estimate of monthly precipitation because it combines multiple satellite data and monthly rain gauge data from the Global Precipitation Climatology Centre (GPCC) (Huffman et al. 2009; Huffman and Bolvin 2017). The multiple satellite data are from passive microwave (PMW) sensors and IR-based (IR) observations, respectively. Specifically, the PMW sensors contain the TRMM Microwave Imager (TMI), the Advanced

Microwave Scanning Radiometer for the Earth Observing System (AMSR_E), the Special Sensor Microwave/Imager (SSMI) and the Special Sensor Microwave Imager/Sounder (SSMIS). And then the IR data are produced by the Geosynchronous Operational Environmental Satellites (GOES), the Geosynchronous Meteorological Satellite (GMS), the Multifunctional Transport Satellite (MTSat) and the Meteorological Satellite (Meteosat). Moreover, the monthly satellite-gauge estimate is computed in three steps process. Firstly, all 3-hourly satellite-only estimates in a month are summed to the monthly value. Secondly, these satellite-only estimates are adjusted at a large-scale using the monthly gauge analysis from Legates and Willmott (1990). Finally, the adjusted satellite estimate is combined with the gauge analysis using inverse error variance weighting. Because the precipitation radar has been out of commission, the climatologically calibrated and adjusted TMPA products have been used since October 2014. Furthermore, a period of transition from the TMPA products, including 3B43V7, will be supported over an appropriate interval to assist users in carrying on TMPA production until mid-2018 (Huffman et al. 2017a).

The GPM mission is the successor of TRMM, which is an international constellation of satellites and consists of one Core Observatory satellite and or so ten partner satellites. Compared to the TRMM instruments, the new capabilities of GPM are the high-frequency channels (165.6 and 183.3 GHz) are added to GMI, and a Ka-band (35.5 GHz) is added to the DPR. The increasing high-frequency channels improve the ability to detect solid precipitation. Therefore, GPM DPR is more sensitive than the TRMM PR in measuring light rainfall and high-latitude snowfall. Furthermore, the DPR is capable of detecting and estimating severe rainfall more accurately, which would be considerable affection on the global climate models so as to better comprehend the global circulation of water (Liu and Zipser 2015). The IMERG data in this study come from the precipitation-estimation algorithm of the Day-1 multi-satellite of GPM (Huffman et al. 2017a), which combines fitful precipitation estimates from multiple constellation microwave sensors [e.g., GMI, AMSR2, SSMIS, the Advanced Microwave Sounding Unit (AMSU) and the Advanced Temperature and Moisture Sounder (ATMS), etc.], IR-based observations (e.g., GOES, GMS, MTSat, Himawari, and Meteosat, etc.) and monthly rain gauge precipitation data (e.g., GPCC products) (Hou et al. 2014; Huffman et al. 2017a, b). Note that the IMERG data provide the probability of liquid phase for all grid boxes, which is a key information to separate estimates for each phase. However, the TMPA only consider all forms of precipitation, which leading to the poor performance in cold season due to its failure to estimate in the frozen or icy regions. In addition, the same to 3B43, the 3IMERGM estimates also are combined with monthly GPCC gauge data. Moreover, given the significant improvements corrected by CPC daily gauge analysis, it will be employed to the future version of IMERG. Three modes (early, late, and final runs) are available with the IMERG product, depending on latency and requirements. The level-3 IMERG final-run products have been provided by the GPM mission since March 2014. Because monthly precipitation gauge analyses are used, the final run provides more accurate results in these regions and is considered to be the research-level product (Huffman et al. 2017a). All three steps produce a 0.5 h, $0.1^\circ \times 0.1^\circ$ resolution, 60°N–60°S spatial coverage dataset (3IMERGHH) and finally yield a monthly $0.1^\circ \times 0.1^\circ$ combination dataset from satellite and gauges (3IMERGM) (Huffman et al. 2017a, b). The latest V05 version of IMERG Final Run product, which has been available since March 2014 and was released in November 2017, is used in this study. Note that since the IMERG data have a different spatial resolution with 3B43V7, a bilinear interpolation method was used to resample the 3B43V7 data so as to obtain the same spatial resolution of $0.1^\circ \times 0.1^\circ$ as IMERG.

Rain gauge data

According to the independence of the evaluation between gauge and satellite precipitation data, there are 194 GPCC stations used for bias adjustment of TRMM/GPM research-level products in China. In particular, there are 18 GPCC stations over Xinjiang (Shen et al. 2013; Tang et al. 2016b). Therefore, we removed the 18 GPCC stations and used the residual 84 stations to the evaluation. Daily precipitation observations from 84 meteorological stations were obtained from the Xinjiang Meteorological Information Center. The spatial distribution of the stations is shown in Fig. 1. It is apparent that the study area has a very uneven distribution of stations. The gauge data used in this study are from April 2014 to August 2017 and have gone through severe quality control procedures. There were no missing records in the monthly time series precipitation data for any of the 84 stations. Monthly precipitation values were accumulated from daily observations to validate GPM and TRMM products. Because the stations are sparsely distributed in Xinjiang and interpolation based on in situ data would result in serious errors, this study used a more direct comparison, i.e., the stations within a grid box were extracted and matched to the box. If two or more stations were located in the same box, the mean precipitation of them was matched with the satellite estimate of the box (Ma et al. 2009; Ward et al. 2011; Blacutt et al. 2015; Hu et al. 2016). It is worth mentioning that the minimum distance among 84 gauges is 11.13 km, which greater than the length of $0.1^\circ \times 0.1^\circ$ IMERG grid. Moreover, the nearest distance two station are not in the same grid box. Therefore, a total of 84 samples (pixels) were evaluated and compared.

DEM

In this study, the Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) DEM data were used to extract terrain variables. The DEM data, with coverage between 83°N and 83°S and a resolution of 1 arc s (30 m horizontal interval at the Equator), were produced based on the detailed observations of the Terra satellite, which is NASA's next-generation Earth observation satellite. In addition, the DEM data were resampled to 500 m × 500 m cell size for the purposes of this study. By using the function of "Aspect," "Slope," "Hillshade," "Tabulate Area" and "Zonal Statistics" of Geographic Information System (GIS) software, we obtained the terrain variables. More details please refer to the literatures. (Yin et al. 2004, 2008).

Methodology

Terrain variables for precipitation correction

The correction factors used in this study included location variables (longitude and latitude) and terrain variables (aspect, slope, roughness, relative relief, etc.), which are commonly considered in studies of orographic effect (Basist and Bell 1994; Roe et al. 2002; Guan et al. 2005; Crochet 2009; Chatzithomas et al. 2015). The location and terrain variables reflect spatial trends and surface conditions such as surface roughness, mountain peaks and ridgelines, and slopes. They are considered to represent potential impacts of precipitation (Yin et al. 2004, 2008). Based on the statistical results of different DEM resolution and different buffers in diameter for each station, the effect corrected by the terrain variables extracted from the 500 m resolution of DEM and a buffer of 40 km is best. Therefore, we choose 500 m resolution DEM data within a buffer of 40 km for each station to further process. In all, 25 factors were extracted (Table 1).

Principal component analysis

Principal component analysis (PCA) is a study of how to translate multiple indicators into less comprehensive indicators, which is a linear combination of the original multiple indexes. These comprehensive indicators are not related to each other and can reflect the information of original multiple variables. In this study, in order to reduce the redundancy in these terrain variables, we conducted PCA by covariance matrix to transform a number of possibly correlated variables into a smaller number of variables. These smaller numbers of variables named principal components (PCs). Firstly, the correlation between in situ measurements and correction factors was analyzed. Secondly, the variables that had a statistically significant correlation (at the 0.05 level) with gauge precipitation were reserved and used for further PCA (Bro and Smilde 2014; Yin et al. 2008). For PCA, similar variables that were deemed to constitute a PC were condensed into a new dataset. In addition, the first PC explains the largest part of the variance in the primitive variables. By removing it, the second PC is obtained, which shows the most variance in the remaining variables and is orthogonal to the first one. This process is continued until the next PC is not significant, and then the redundant variables in the original dataset are removed (Yin et al. 2008). In this study, for each terrain variable, there were 84 samples, one for each rain gauge. The varimax method, which is the most popular rotation method, was used to rotate the resulting PCs, but with a "simple structure" rotation (Kaiser 1958; Yin et al. 2008). It is a commonly used method in principal component analysis or factor analysis. After coordinate transformation, the sum of variance of each factor load is maximum. That is, any one of the variables has a high loading value (high contribution rate) only in one factor or a small number of variables, and the loading value is almost zero in other factors. The factor loading matrix satisfied above condition are called "simple structure". The eigenvalue of each PC was calculated based on varimax rotation. Finally, factors with similar effect on precipitation were classified as one RPC and calculated according to the RPC scores of each station for the further analysis.

The specific calculation process is as follows:

1. The matrix X of the standardized data index was obtained. The n columns are the number of rain gauges ($n = 84$), and the m rows are the terrain factors ($m = 25$).

$$X = \begin{pmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots \\ x_{m1} & \cdots & x_{mn} \end{pmatrix} \tag{1}$$

2. Compute the covariance matrix all the possible pairs between the m variables.

$$S = (s_{ij}) = \frac{1}{n}\sum_{t=1}^{n} \begin{pmatrix} x_{1t} - \overline{x_1} \\ \vdots \\ x_{mt} - \overline{x_m} \end{pmatrix} \cdot \begin{pmatrix} x_{1t} - \overline{x_1} \\ \vdots \\ x_{mt} - \overline{x_m} \end{pmatrix}^{\mathrm{T}} \tag{2}$$

where s_{ij} is the covariance between the i factor and the j factor ($i = j$=1,2, …m). Note that S is an $m \times m$ symmetric matrix.

3. The eigenvalue of the covariance matrix, i.e., the variance contribution rate of each PC, and the corresponding eigenvector are calculated. After that, the first q principal components with the eigenvalue greater than 1 and the cumulative variance contribution rate greater than 85% are selected.

Table 1 Definition and description of terrain variables

Name	Description
MINele	Minimum elevation within 40 km buffers
MAXele	Maximum elevation within 40 km buffers
RANGEele	Range of elevation within 40 km buffers
MEANele	Mean elevation within 40 km buffers
STDele	Standard deviation of elevation within 40 km buffers
SUMele	Sum of all elevation within 40 km buffers
MEDINele	Median elevation within 40 km buffers
MAXslp	Maximum slope within 40 km buffers
MEANslp	Mean slope within 40 km buffers
RANGEslp	Range of slope within 40 km buffers
STDslp	Standard deviation of slope within 40 km buffers
SUMslp	Sum of slope within 40 km buffers
MEANhshd	Mean lighting condition within 40 km buffers
min_h	Minimum relative relief within 40 km buffers
max_h	Maximum relative relief within 40 km buffers
mean_h	Mean relative relief within 40 km buffers
std_h	Standard deviation of relative relief within 40 km buffers
n1_asp	Area percent of north-facing slopes within 40 km buffers
ne2_asp	Area percent of northeast-facing slopes within 40 km buffers
e3_asp	Area percent of east-facing slopes within 40 km buffers
se4_asp	Area percent of southeast-facing slopes within 40 km buffers
s5_asp	Area percent of south-facing slopes within 40 km buffers
sw6_asp	Area percent of southwest-facing slopes within 40 km buffers
w7_asp	Area percent of west-facing slopes within 40 km buffers
nw8_asp	Area percent of northwest-facing slopes within 40 km buffers

All variables are obtained from the DEM data

4. Take the corresponding q eigenvectors and standardize them. The loading value (w_q) of each original variable in each eigenvector was sorted in descending order and recombined.
5. Calculate the score (F) of each principal component:

$$F_q = w_{qm} \cdot x_{mn} \tag{3}$$

where w_{qm} is the loading value of the qth PC for each factor and x_{mn} is standardized data index.

Stepwise regression method

The basic idea of stepwise regression is to enter variables into the model one by one. When the introduced original explanatory variable is no longer significant due to the introduction of the other explanatory variable, it will be removed, to ensure the regression equation only contains significant variables before every new variable introduced. This is an iterative process until no significant explanatory variables are selected and significant explanatory variables are excluded from the regression equation, which ensure the final data set of explanatory variables is optimal.

Stepwise regression was performed to acquire a multiple regression equation. In this study, monthly IMERG estimates, RPCs, and location variables were used as the independent factors and corresponding in situ measurements as the dependent factor. Stepwise regression was considered a reasonable method to use because it may be treated as a search procedure to recognize which independent variables are useful and to find the strongest relationship between independent and dependent variables to derive a "best" equation (Beale et al. 1967; Newton and Spurrell 1967). It also was well suited for the study area because of the complex terrain in Xinjiang.

In addition, longitude, latitude and terrain variables were extracted from the 500 m resolution DEM within a buffer 40 km in diameter for each grid cell of the 3IMERGM data, and the regression model was applied to each grid cell of the 3IMERGM precipitation estimates. Finally, corrected areal precipitation with 0.1° × 0.1° monthly resolution for the entire study region was produced.

Evaluation metrics

To evaluate the satellite precipitation products comprehensively across Xinjiang, three metrics were selected: correlation coefficients (CC), relative bias (RB), and root mean square error (RMSE). The formulas used to calculate these metrics are:

$$\mathrm{RB} = \frac{\sum (x_i - y_i)}{\sum y_i} \times 100\% \tag{4}$$

$$\mathrm{CC} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} \tag{5}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n} \sum (x_i - y_i)^2} \tag{6}$$

where x and y are satellite precipitation estimates and in situ measurements respectively, $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$, $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$ represent the mean values of these variables, and n is the number of all matched monthly precipitation samples.

In addition, using the modeling process described above, the tenfold cross-validation method was used to test the accuracy of the model. First, all valid data were divided into ten parts at random, nine of which were taken as training data and one as test data. This process was repeated ten times, after which 12 regression models corresponding to the 12 months of the year were established by the modeling method for each test. Each site was then verified independently at the end of this process. By collecting the records of each test, complete data sets with corrections were acquired. All the following evaluations of C-3IMERGM are based on this dataset.

Results

Evaluation of 3B43V7 and 3IMERGM

Figure 2 shows the monthly estimated values of GPM 3IMERGM and TRMM 3B43V7 plotted against gauge values. As shown in the figure, 3IMERGM precipitation estimates showed better correlations with ground observations than 3B43V7. The coefficients of determination for 3IMERGM and 3B43V7 were 0.47 and 0.39, respectively. It is noted that TMPA exhibits anomalous overestimations over inland water bodies due to the limitations of passive microwave retrievals (Tang et al. 2016a). Therefore, the overestimation of weather stations in these waters is one of the reasons for the low quality of TMPA precipitation assessment. Table 4 shows a slight decrease in RMSE for 3IMERGM compared to 3B43V7 (19.02 mm for 3B43V7 and 17.42 mm for 3IMERGM). As for RB, both products showed overestimation overall, but the improvement of 3IMERGM over the TRMM3B43 estimates is obvious, with RB values decreased from 10.24 to 7.76%.

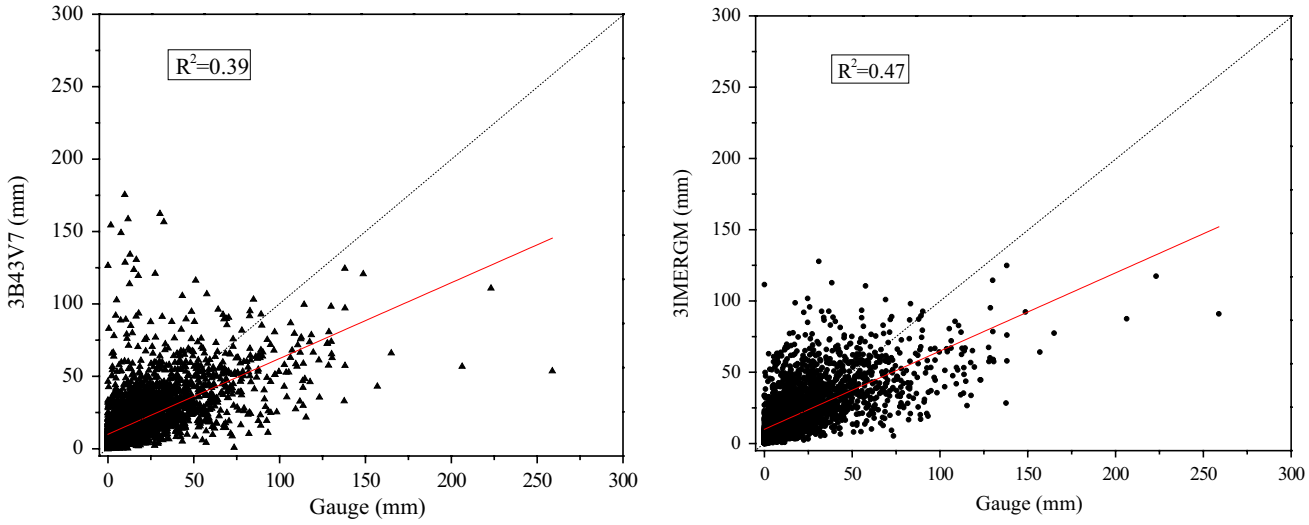


Fig. 2 Scatter plot of 3IMERGM and TRMM 3B43V7 against observed monthly precipitation values at all stations from April 2014 to August 2017 over Xinjiang (the dot lines and red lines are 1:1 lines and fit lines, respectively)

Based on the monthly precipitation spatially averaged from all stations, Fig. 3 shows a time series of the multiple data sets from April 2014 to August 2017. The 3IMERGM precipitation is closer to rain gauge data, especially after January 2015, while similar to the 3B43V7 before 2015. Figure 3 also shows that the 3B43V7 product overestimated monthly precipitation in most months and that the 3IMERGM product performed better than 3B43V7, especially in the winter season (December to the following February). From the records in winter during the study period, the statistics of these records showed that the RBs for 3B43V7 and 3IMERGM were 52.3 and 20.47%, respectively. It is obvious 3B43V7 over twice as high than 3IMERGM, which shows solid precipitation was greatly overestimated by 3B43V7 in winter. The box plot in Fig. 4 also shows that 3B43V7 overestimates precipitation more than 3IMERGM in the winter season. There are two main reasons leading to this phenomenon: first, passive microwave sensors were influenced by seasonal snow–ice cover in this study area due to misclassify them as rain clouds (Hirpa et al. 2010), thus leading to precipitation false alarms and overestimation; second, the raindrops evaporated before it reached the ground, is a possible reason leading to these overestimations (Ferraro et al. 1998).

Figure 4 shows the absolute error of monthly precipitation between 3IMERGM, 3B43V7, and rain gauge data for each month for all stations together. TRMM 3B43V7 overestimated the in situ measurements during the cold season

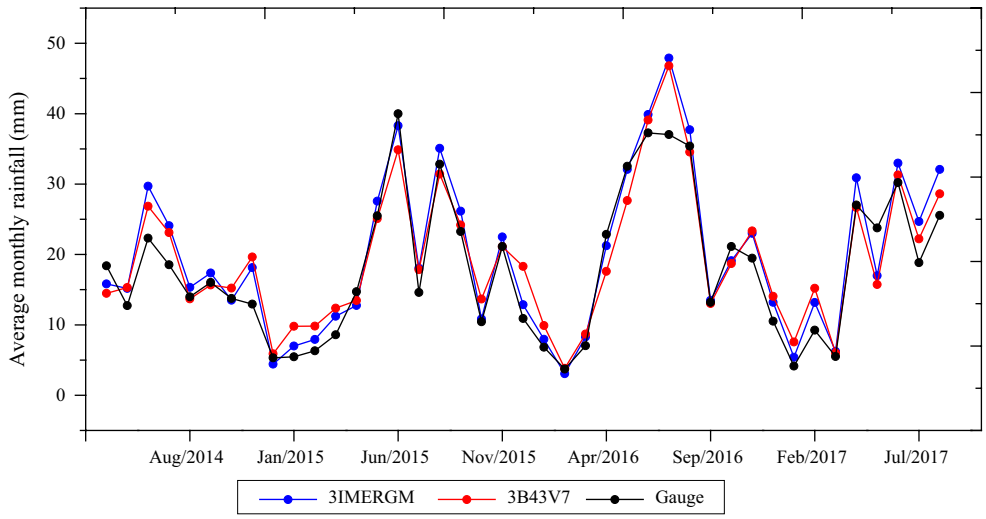


Fig. 3 Time series of monthly precipitation (mm) between satellite estimates and rain gauge data for all stations from April 2014 to August 2017 over Xinjiang

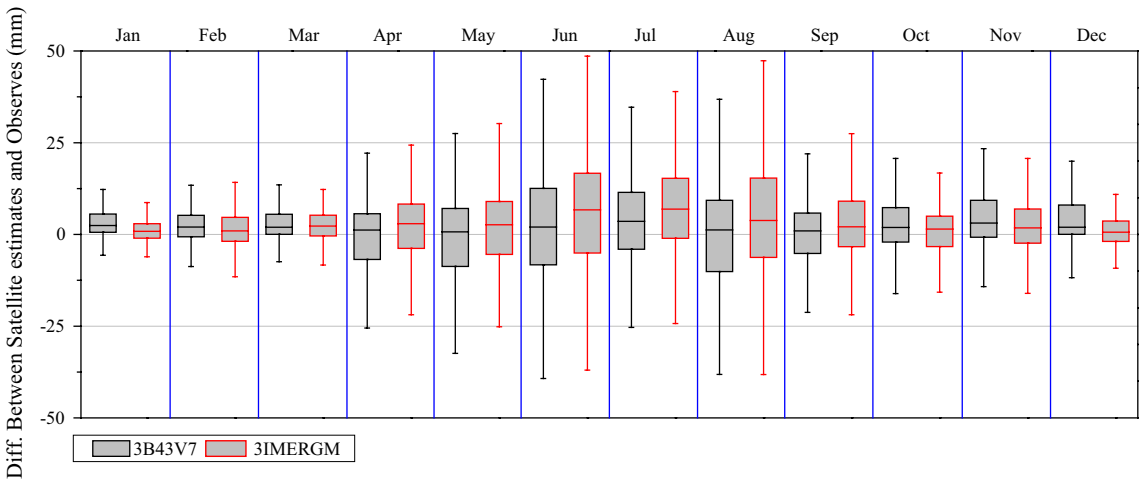


Fig. 4 Box plot of monthly variations of absolute error between satellite estimates and rain gauge data for all stations from April 2014 to August 2017 over Xinjiang

(October to the following April); however, a significant improvement was achieved by 3IMERGM for the same period. The distribution of 3IMERGM was uniform on both sides of the zero-horizontal line except for March. During the warm season (May to September), 3B43V7 overestimated station precipitation slightly, but the overestimation range of 3IMERGM estimates against gauge data was larger than 3B43V7 during the same period. In general, from the absolute error, 3IMERGM datasets performed better than 3B43V7 during the cold season, but slightly worse than 3B43V7 during the warm season. A possible reason is GPM extends the TRMM sensors and enhances the capacity of observing light and solid precipitation (Huffman et al. 2017a). Thus, differences in the observation ability compared with the large precipitation in the tropical and subtropical observation on TRMM, GPM can more accurately capture the light precipitation (< 0.5 mm/h) and solid precipitation, while the two type of precipitation is an important part of the precipitation in middle-high latitudes, e.g., the Xinjiang region.

Correction of the 3IMERGM product

Understanding satellite precipitation errors and their random and systematic components is essential to correct the satellite precipitation (AghaKouchak et al. 2012; Maggioni et al. 2016). In terms of the method developed by Willmott (1981), the systematic and random error components (%) of 3IMERGM product in our study are 37.8 and 62.2, respectively, which indicated the systematic error is significantly less than random error at the monthly time scale due to the GPCC stations adjustment performed in the 3IMERGM research product. Note that the systematic error still large because of scarce GPCC stations coverage (only 18 gauges) (Habib et al. 2004; Maggioni et al. 2016). Therefore, we use more rain gauges and combined with topographic factors to further correction for 3IMERGM.

Twenty-five terrain variables that had a potential impact on the precipitation distribution were extracted from the DEM data (Table 1). A selection was then made among those variables by correlation analysis to retain the variables that had statistical significance in forecasting the spatial distribution of precipitation. Correlation analysis was performed using terrain variables against gauge data when total data were gathered during the study period. Twenty-three factors with a significant correlation (at the 0.05 level) with gauge data were reserved to conduct the subsequent RPC procedure, leaving aside *ne2_asp* and *sw6_asp* (Table 2). After performing PCA, four rotated PCs (RPCs) were determined, which explained over 86% of the variance in the original dataset. Each RPC contained those terrain variables that reflected the loading values from the most negative to the most positive and represented at least one initial terrain variable. Each RPC consisted of the terrain variables listed in Table 2, which indicated the important of topographic features in enhancing satellite estimates (Yin et al. 2008). For these RPCs, RPC1 represents mostly variability of elevation or surface roughness; RPC2 represents slope aspect, especially the northwest-facing and north-facing slopes toward windward slope (positive correlation), while southeast-facing and south-facing slopes toward leeward slope (negatively correlated) in Xinjiang; RPC3 is

Table 2 Results of rotated principal component analysis with eigenvalues for each RPC and topographic variable

RPCs		RPC1		RPC2		RPC3		RPC4
Eigenvalues		11.722		3.438		2.796		1.49
	Variable	Loading value	Variable	Loading value	Variable	Loading value	Variable	Loading value
Represented variables	MAXele	0.954	n1	0.867	MINele	0.801	w7	0.841
	STDslp	0.954	nw8	0.845	MEDIANele	0.612	e3	− 0.654
	STD_h	0.951	s5	− 0.792	MEANele	0.568		
	MEAN_h	0.945	se4	− 0.852				
	MEANslp	0.9						
	RANGEele	0.886						
	MAXslp	0.885						
	RANGEslp	0.885						
	MAX_h	0.872						
	SUMslp	0.864						
	STDele	0.858						
	SUMele	0.646						
	MEANhshd	− 0.717						
	MIN_h	− 0.910						

a measure of elevation overall associated with the forced ascent of topographic forcing. The size of the eigenvalue represents contribute rate of this PC to the entire matrix. Scores of the four RPCs were used in the stepwise regression procedure to correct 3IMERGM precipitation estimates.

The 3IMERGM estimate was treated as a forcing factor in each process, i.e., the necessary factor in the process of establishing equations, but all the other variables were treated as conditional factors by the stepwise regression procedure. The stepwise regression approach was conducted using monthly IMERG estimates, the RPCs, and location variables as the independent variables and monthly gauge data as the dependent variable. Table 3 is the standard regression coefficient of stepwise regression analysis after normalizing the independent variables and dependent variables. Table 3 shows that the constant of the standard regression equation is 0. In addition, Table 3 also exhibits 3IMERGM explained the variance of dependent variable is smaller than other months in January and February, while Lat and Lon both provided more contribution rate than 3IMERGM. In other months, it is basically 3IMERGM explained about 40% of the dependent variable, i.e., rain gauge data. This reflects the quality of IMERG needs to be improved in winter. The most frequent RPCs included in the final models in stepwise regression are RPC2 (12 times), which indicates the slope aspect has an influence on rain gauge data each month, followed by RPC1 (8 times), RPC3 (5 times), and RPC4 (4 times). Table 3 shows the longitude and latitude are basically influenced on gauge data in the cold season, and the gauge data increased with the increase of latitude, while decreased with the increase of longitude. RPC1, represents mostly variability of elevation or surface roughness, explained over 30% of the dependent variable in the warm season (May to September). The above reflect the Lon, Lat and RPC1 present distinct seasonal variations. Finally, 12 equations were defined (Table 3) by applying these equations to each grid cell of the 3IMERGM data, eventually producing the C-3IMERGM dataset.

Comparison between C-3IMERGM, original GPM IMERG and TRMM 3B43 datasets

Assessment over all Xinjiang

In this work, the CC, RB, and RMSE were used to verify the accuracy of the satellite estimates and the corrected data over the whole study area. The accuracy of the satellite products was evaluated by comparing in situ measurements with retrieved precipitation at each station on a monthly scale. Overall, the 3IMERGM product performed better than 3B43V7 in the study period. However, the data after correction by the terrain factor were the best among the three data sets. In particular, CC and RB showed obvious improvement (Table 4).

To further reflect the distribution of the evaluation metrics, the CC, RB, and RMSE are shown by box plots in Fig. 5 over the domain between gauge data and retrieved precipitation for all stations pooled together. Overall, the median CC of C-3IMERGM was higher (0.79) than the other two products with 0.71 (3B43V7) and 0.77 (3IMERGM). In particular, the distributions of RB for C-3IMERGM showed better symmetry, with values close to zero, and the values of the first/third quartiles were the lowest among the three

Table 4 Mean values of CC, RB (%), and RMSE (mm) for 3B43V7, 3IMERGM, and corrected precipitation estimates at monthly timescale from April 2014 to August 2017 for all Xinjiang

	CC	RB (%)	RMSE (mm)
3B43V7	0.62	10.24	19.02
3IMERGM	0.68	7.76	17.42
C-3IMERGM	0.73	− 1.65	16.13

Table 3 Results of stepwise regression analysis for 3IMERGM for each month

Month	Constant	3IMERGM (mm)	Lon (°E)	Lat (°N)	RPC1	RPC2	RPC3	RPC4
Jan	0	0.376	− 0.413	0.189		0.239	− 0.308	0.232
Feb	0	0.375	− 0.264	0.17	0.099	0.344	− 0.23	0.173
Mar	0	0.635			0.233	0.119		
Apr	0	0.575		0.088	0.299	0.220		
May	0	0.574			0.325	0.226		0.11
Jun	0	0.573			0.327	0.239		0.152
Jul	0	0.618		0.106	0.358	0.148	0.178	0.103
Aug	0	0.541			0.421	0.191	0.1	
Sep	0	0.642			0.268	0.233		0.144
Oct	0	0.713			0.111	0.197		0.09
Nov	0	0.366	− 0.323	0.318		0.236	− 0.178	0.222
Dec	0	0.526	− 0.197	0.172		0.245	− 0.198	0.184

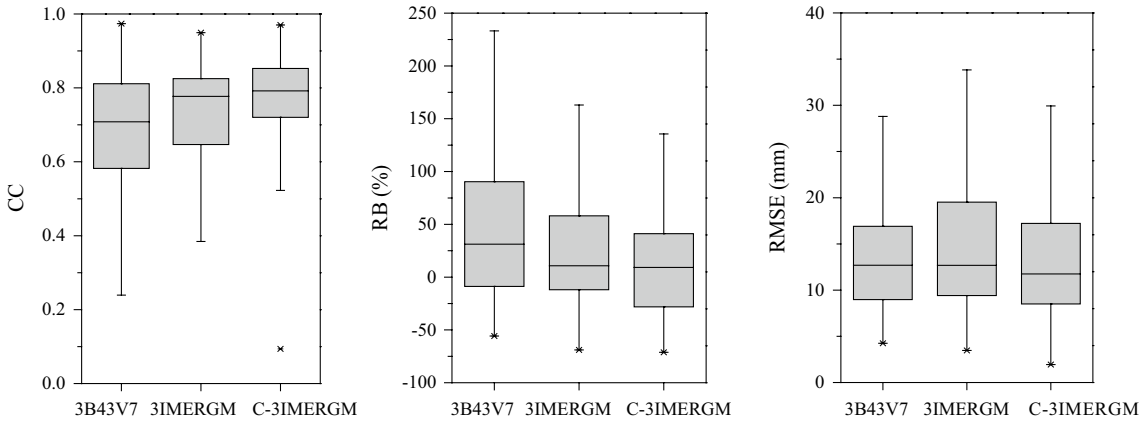


Fig. 5 Box plots for 3B43V7, 3IMERGM, and C-3IMERGM precipitation data at monthly timescale

data sets. RMSE showed similar performance to RB, with the lowest errors (11.75 mm) found for C-3IMERGM. In contrast, the RMSEs of 3B43V7 and 3IMERGM were 12.69 and 12.68 mm, respectively. From this analysis, it can be concluded that the corrected precipitation performs better and is acceptable at the monthly scale over all Xinjiang.

C-3IMERGM precipitation estimates were then compared with rain gauge data to assess further the effectiveness of the regression method. Figure 6 shows CC, RB, and RMSE for each month from April 2014 to August 2017 for the three datasets. The CC of 3IMERGM was higher than 3B43V7 for 10 months and lower than 3B43V7 for 2 months. For the C-3IMERGM data, except for March and June, which were slightly lower than the original 3IMERGM product, the other months all improved. The improvement obtained by C-3IMERGM is more clearly reflected by RB and RMSE. Figure 6 illustrates that RB and RMSE were improved for most months over the original IMERG precipitation estimates; the RMSE after correction ranged from 4.95 to 26.35, decreasing by 0.11–2.61 mm from previous values.

This analysis indicates that the C-3IMERGM data performed better than the original precipitation-estimation products overall, which implies that the 3IMERGM data were improved by the stepwise regression method and that the corrected GPM 3IMERGM can be accepted.

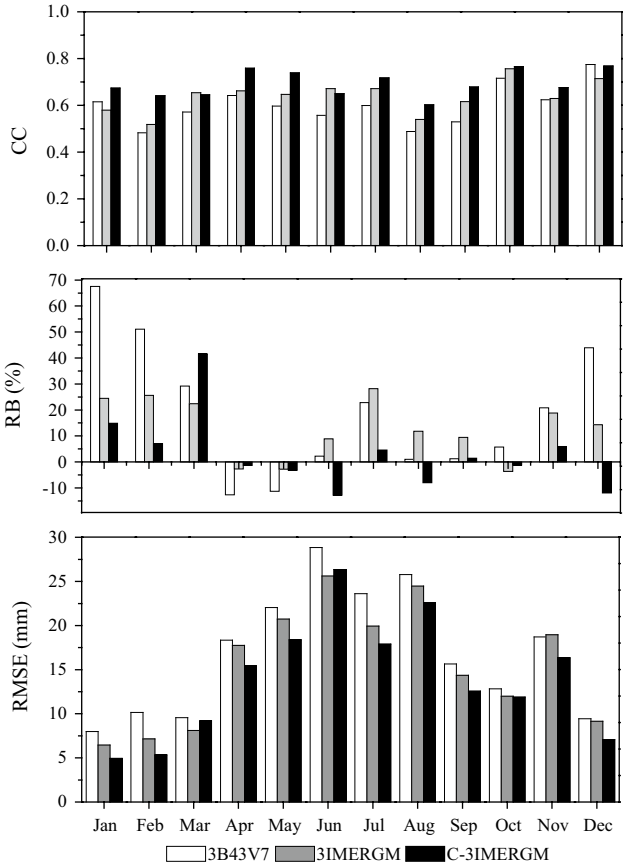


Fig. 6 Histogram for 3B43V7, 3IMERGM, and C-3IMERGM precipitation estimates products at monthly timescale

Elevation impact analyses

All the sites in Xinjiang were rearranged in order of increasing altitude, ranging from − 47.4 to 3507.2 m (Fig. 7). For different station altitudes, Table 5 shows a comparison of the CC, RB, and RMSE between satellite estimates and rain gauge data at monthly scale. For the elevation analyses, the entire Xinjiang area was separated into three elevation zones: < 1000, 1000–1500, and > 1500 m. Based on this classification, an evaluation was conducted for all the gauge stations from April 2014 to August 2017 at monthly scale. For mean precipitation, the three datasets showed the same trend: with increasing altitude, precipitation first decreased and then trended upward. Note that 3IMERGM significantly underestimated precipitation at over 1500 m. The CCs of

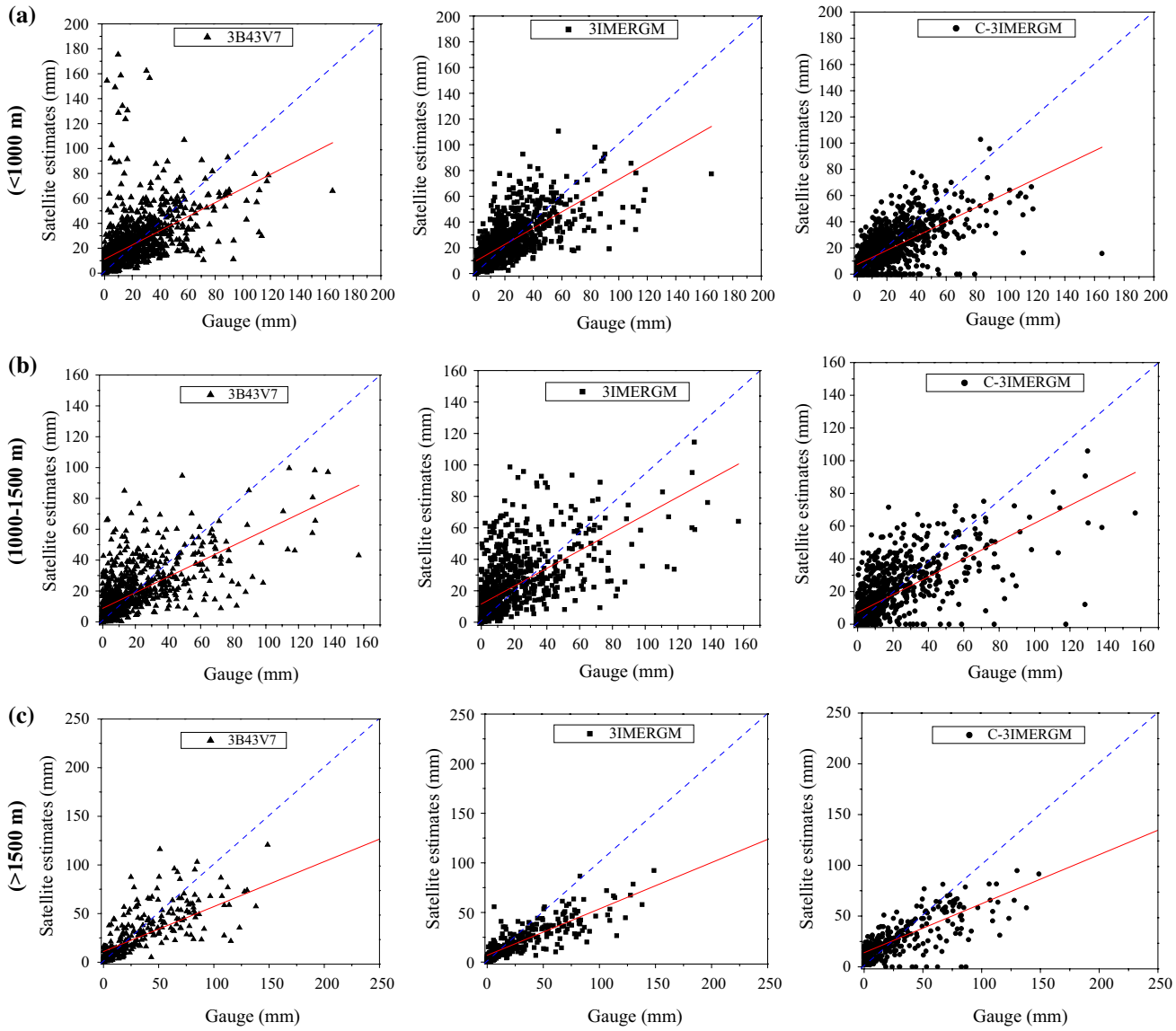


Fig. 7 Scatter plots of monthly precipitation for in situ measurements, satellite precipitation estimates, and C-3IMERGM in the three elevation zones: **a** less than 1000 m, **b** 1000–1500 m, **c** greater than 1500 m (the dash lines and red lines are 1:1 lines and fit lines respectively.)

the 3IMERGM data were all higher than those for 3B43V7 in the three altitude zones and were further improved after correction. For RB, the three datasets all overestimated precipitation in the first two zones, but underestimated it in the third zone (over 1500 m). For RMSE, the maximum error occurred at elevations over 1500 m.

After correction, the statistics showed that the correction method performed best for 3IMERGM in lowland areas (under 1000 m), with the best correlation (0.73), minimum RB (4.31%), and lowest RMSE (11.76 mm) of the three altitude zones. For the 1000–1500 m zone, the increment (6%) in CC is the most obvious. RB was higher (6.85%), meaning that C-3IMERGM overestimated precipitation in this altitude zone. At elevations over 1500 m, the CC between C-3IMERGM and rain gauge data was 0.7, RB was − 5.95%, and RMSE increased to 23.43 mm, indicating that underestimation remained in high-latitude areas even after correction. Overall, the C-3IMERGM precipitation estimates performed better on the indicators than the unadjusted data.

Regional analyses

For the regional analyses, the whole Xinjiang area was separated into two parts by the Tianshan Mountains. The two parts were the SX zone and the NX zone (Fig. 1). Previous studies have shown that NX and SX have different climatic regimes (Li et al. 2016; Tan and Shao 2016). In terms of temperature, the northern area is colder than the southern

Table 5 Statistical results for in situ measurements, satellite precipitation estimates, and C-3IMERGM in three elevation zones

	< 1000 m	1000–1500 m	> 1500 m
No. of stations	37	33	14
Mean (mm)			
Observed	17.98	12.85	30.47
3B43V7	21.48	14.79	25.07
3IMERGM	21.26	17.87	21.07
C-3IMERGM	17.21	13.73	28.67
CC			
3B43V7	0.55	0.60	0.68
3IMERGM	0.68	0.63	0.69
C-3IMERGM	73	0.69	0.7
RB (%)			
3B43V7	19.43	39.07	− 17.74
3IMERGM	18.22	15.11	− 30.86
C-3IMERGM	4.31	6.85	− 5.95
RMSE (mm)			
3B43V7	18.49	15.98	25.77
3IMERGM	14.04	17.75	24.06
C-3IMERGM	11.76	14.42	23.43

region. The mean temperatures of NX in winter and summer are − 13 and 22.2 °C, respectively, with a continental arid and semiarid climate, whereas those for SX are − 5.7 and 24.4 °C, respectively, with a continental dry climate (Li et al. 2011a). In this study, the SX and NX zones had 40 and 44 weather stations, respectively. Figure 8 shows scatter plots of the stations for the two parts, and Table 6 presents the results of their evaluation accordingly. Results showed that precipitation in NX was two times greater than in SX. The CCs of the three datasets in SX were all higher than in NX. On the contrary, the RBs in NX were all less than in SX. However, for RB, the data correction showed a negative effect in NX, but a positive effect in SX. Finally, the RMSEs in SX were all lower than in NX. This analysis indicates a need for further improvement of the correction method in NX. Judging from the performance of the three products, TRMM 3B43V7 overestimated precipitation in both zones, especially in SX. 3IMERGM underestimated precipitation in NX, but overestimated it in SX. As for C-3IMERGM, the degree of overestimation was reduced in SX to some extent, but the underestimation (− 6.65%) was slightly greater than with the original 3IMERGM (− 4.68%) in NX. Nevertheless, the overall degree of underestimation was less than 10%, and the CC and RMSE of C-3IMERGM in both SX and NX were both greatly improved.

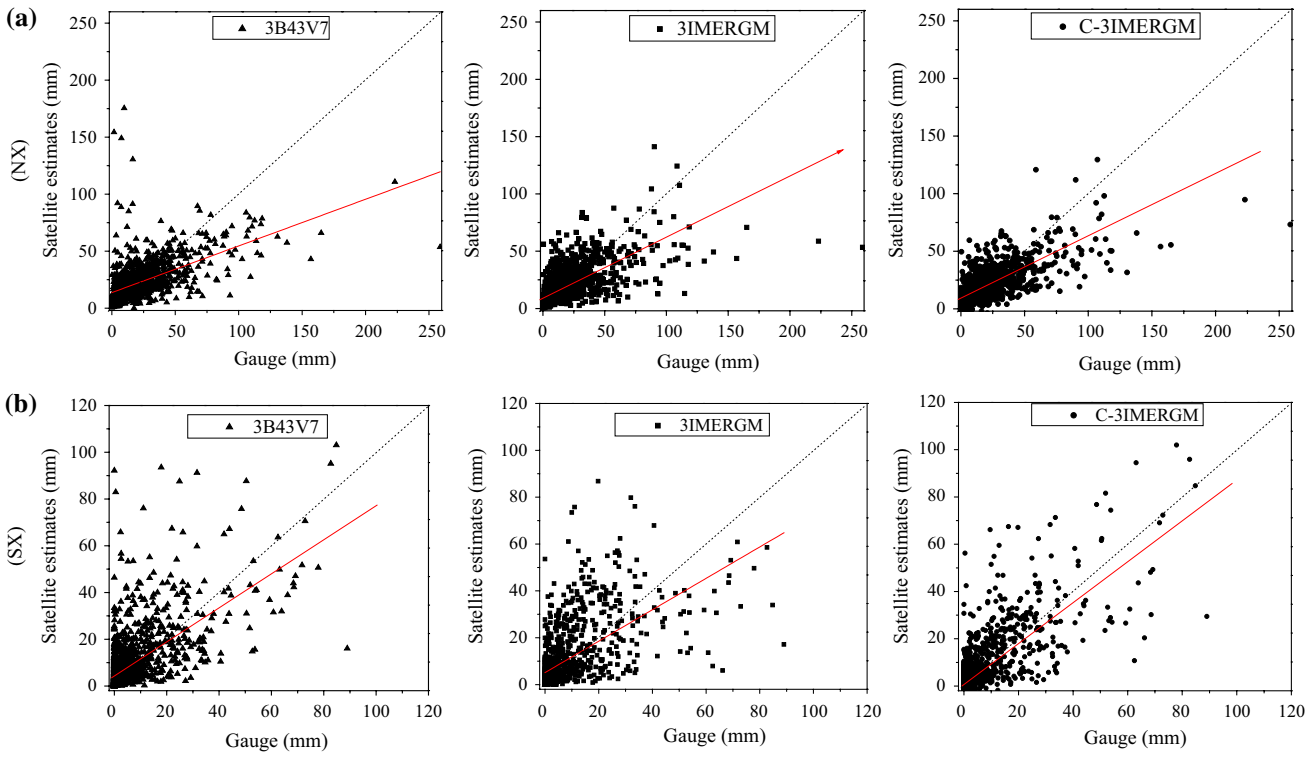


Fig. 8 Scatter plots of monthly precipitation for in situ measurements, satellite precipitation estimates, and C-3IMERGM in the two zones: **a** NX, **b** SX (the dot lines and red lines are 1:1 lines and fit lines, respectively)

Table 6 Statistical results for in situ measurements, satellite precipitation estimates, and C-3IMERGM for the two zones in Xinjiang

	NX	SX
No. of stations	40	44
Mean (mm)		
Gauge	26.67	10.21
3B43V7	25.58	13.87
3IMERGM	25.42	14.87
C-3IMERGM	23.3	12.71
CC		
3B43V7	0.53	0.67
3IMERGM	0.60	0.65
C-3IMERGM	0.7	0.72
RB (%)		
3B43V7	4.09	45.69
3IMERGM	− 4.68	35.93
C-3IMERGM	− 6.65	24.5
RMSE (mm)		
3B43V7	23	14.47
3IMERGM	19.26	15.56
C-3IMERGM	19	12.96

Actual spatial pattern of precipitation revealed by C-3IMERGM

The analysis described above as well as the tenfold cross-validation indicated that the correction method has a certain effect on satellite precipitation products. Therefore, longitude, latitude, and terrain variables were extracted from the 500 m resolution DEM for a 40-km diameter buffer for each grid cell of the 3IMERGM data. The optimal model was obtained using the correction methods based on all gauge stations rather than the 90% used for modeling because the greater the number of samples used for modeling in the local area, the better was the model obtained. The optimal model was applied to each grid cell of the 3IMERGM precipitation estimates, and finally areal precipitation estimates with $0.1° \times 0.1°$ monthly resolution for the entire study region were produced.

All the GPM 3IMERGM monthly precipitation-estimation grid cells for 2015 in the study area were corrected using the optimal regression model. Figure 9 represents the spatial distribution of annual precipitation based on 3B43V7, 3IMERGM, and C-3IMERGM in 2015 over Xinjiang. The spatial variations of precipitation in mountainous areas are more complex than in lowland areas. As shown, the precipitation distributions from the three products were consistent in their overall trends, but the 3B43V7 product reflected only rough precipitation distribution trends over the entire study area due to its poor resolution (0.25°). With the higher resolution (0.1°) of the 3IMERGM and C-3IMERGM products, the finer characteristics of precipitation distribution became apparent, an improvement with great significance for many other research studies. Figure 9c also reveals abundant precipitation in the Kunlun Mountains in southern Xinjiang, with an average altitude of more than 5000 m, according to C-3IMERGM precipitation estimates, but the first two products (Fig. 9a, b) showed very little data for this region. In addition, Fig. 9c shows there are rich precipitation in the Yili Valley, which located in the western part of the Tianshan Mountains in Xinjiang. This did not show in the first two products too. The terrain features of Yili Valley are surrounded by mountains on the three sides and opened to the west to form a trumpet shaped Valley, which receive the water vapor of the Atlantic, and the warm and humid currents rise in the windward slope to form orographic rain. Yili Valley has mild and humid climate, abundant precipitation and lush grassland. The annual rainfall is 417.6 mm, and the mountain area is 600 mm. It is the wettest area in Xinjiang. Because of the scarce stations, the areal precipitation generated by the interpolation method will bring great errors, so we can only use the existing separate stations to compare and evaluate the satellite precipitation. The average precipitation values estimated by the three products (3B43V7,

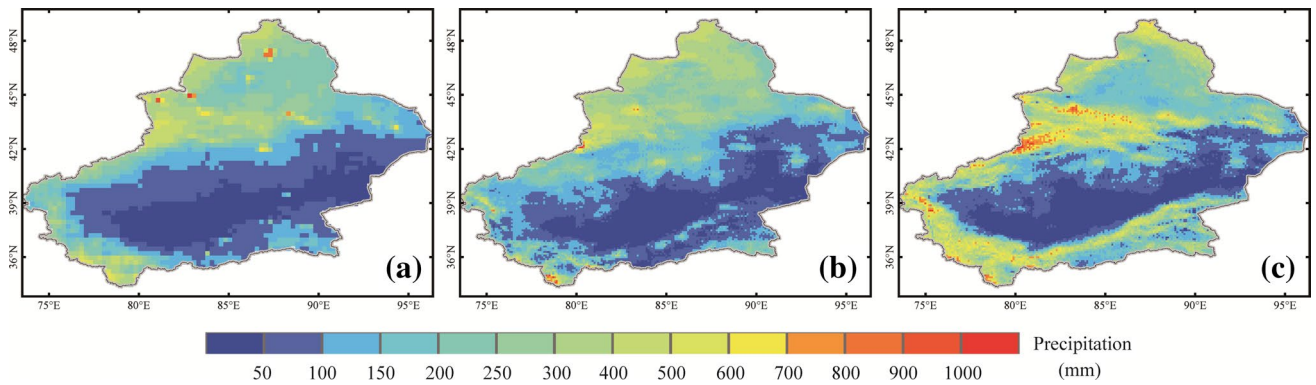


Fig. 9 Spatial distribution of annual precipitation in 2015 over Xinjiang for **a** 3B43V7, **b** 3IMERGM, and **c** C-3IMERGM

3IMERGM, and C-3IMERGM) over the whole study region in 2015 were 175.3, 169.6, and 226.9 mm, respectively, and the maximum precipitation values predicted by 3IMERGM and C-3IMERGM were 1333.6 and 1370 mm, respectively, compared to only 717.6 mm predicted by 3B43V7. From observations, the maximum annual precipitation in 2015 was 893.5 mm at Tianchi station, but this was just the maximum annual precipitation that could be observed at a weather station. These results indicate that TRMM 3B43V7 underestimated heavy precipitation. This occurred mainly because the TRMM core observatory can cover only a small part of Xinjiang, and therefore, the TRMM 3B43V7 product in Xinjiang is mainly derived from combined microwave-IR-gauge estimates of precipitation. In contrast, the GPM core observatory can cover latitudes 65°N–65°S, and therefore its precipitation inversion was uniformly calibrated by its core satellite. Moreover, Fig. 9c shows a low-precipitation area in northern Xinjiang because the area includes the Junggar Basin, which contains the second largest desert (the Gurbantunggut Desert) in China, covering an area of 45,000 square kilometers and second only to the Taklimakan Desert. This was the cause of the low-precipitation area in Fig. 9c. All these results demonstrate that the C-3IMERGM product is more credible than the others studied.

Discussion

Xinjiang is characterized by complex topography. The complex topography, to a great extent, affects the distribution of water resources. At present, much research about the areal precipitation and water resource distribution over Xinjiang is mostly based on the rain gauge data (Li et al. 2011a, b, 2016; Zhao et al. 2011; Zhang et al. 2012, 2015; Xu et al. 2016). Because the weather stations are sparse and mostly built in low altitude and human settlements, this makes them have limitations in characterizing the spatial and temporal distribution of precipitation in the entire Xinjiang region as well as high altitudes area. On the other hand, the performance of satellite estimates precipitation products is still unsatisfactory (Tang et al. 2016b; Huang et al. 2016). Thus, its applications are still at the experimental stage, especially for the mountainous area (Shige et al. 2013).

Areal precipitation is an important input parameter for hydrologic model, numerical weather forecast, climate model and climate diagnosis research. (Karavitis and Kerkides 2002; Chatzithomas et al. 2015). In Xinjiang, the large arid area of China, there are two main methods to obtain areal precipitation. One is interpolating the scarce weather stations by GIS method (Xu et al. 2016; Li et al. 2011a), the other is based on the statistical relationships between terrain factors and rain gauge data (Basist and Bell 1994; Chatzithomas et al. 2015). This paper focuses on the further correction of satellite estimates by taken satellite precipitation as initial areal precipitation and introduced terrain factors. Although the topographic effect has already been considered by the IMERG precipitation retrieval algorithm (Huffman et al. 2017a, b), the satellite rainfall products could be obtained further correction based on the relationship between terrain factors and local rain gauge precipitation. Consequently, our research both contains the achievements of GPM IMERG research-level products and considers the local terrain factors, which can be expected to obtain more accurate areal precipitation over Xinjiang.

The correction effect of terrain factors on satellite precipitation product, greater time scale such as monthly or yearly tend to provide better result than smaller time scale such as hourly or daily (Hutchinson 1973; Yang and Luo 2014a; Chatzithomas et al. 2015). For short time scale, the weather conditions should be considered as the main factors. This can be explained weather condition plays a leading role for the smaller time scale. While for the monthly or yearly time scale, the cumulative effect from topographic influence on precipitation will be highlighted, i.e., the portion of precipitation variability that can be explained by terrain factors becomes bigger with the increase of time scale. This further confirms the validity of correcting satellite estimates monthly precipitation by the topographic factor in this study.

Due to the scale effect of geographic data, the terrain variables values extracted from different resolution DEM and buffers of the weather stations are different. Taniguchi et al. (2013) indicated the optimal horizontal length scale of for averaging the elevation data is about 50 km. In this study, in order to clarify which group of terrain factors to the correction was large, the initial 30 m resolution DEM was resampled to 50, 100, 150, 200, 300, 500, 1000, 1500 and 2000 m, then the buffer is set to ten different thresholds at 5 km intervals (5, 10, 15, …, 50 km), then 100 groups of terrain factors were extracted to establish different equations; finally, the correlation coefficient matrix after the correction was obtained.

The correlation coefficient matrix indicated that the influence to the corrected result from different buffers is greater than that from different resolution. CCs represent a trend from small to large then to small with the increase of buffers, and the maximum value all distribute near the buffer of 40 km. The result is similar to that of Taniguchi et al. (2013). The results exhibit the correction effect of 5 km buffer in each resolution is worst. Furthermore, the maximum and minimum of CC is 0.73 and 0.66, located in 500 m resolution buffer of 40 km and 1500 m resolution buffer of 5 km, respectively. It is shown that the differences of extracted topographic factors values have a great influence on the correction results.

Overall, it is noted that Xinjiang has large area and complex terrain, while taken the entire Xinjiang region as the

study area is not suitable for each subregion. The next work is to divide the entire Xinjiang region into different subregions according to different surface types and to establish different correction model for each subregion so as to improve the explain ability of correction model, as well as distinguish the orographic/nonorographic precipitation (Shige et al. 2013), which is expected to further analyze the topographic effects on precipitation.

Conclusions

This study first evaluated the performance of the TRMM 3B43V7 and GPM 3IMERGM datasets in Xinjiang, China, from April 2014 to August 2017. Then, the 3IMERGM precipitation product was corrected by adding location and terrain variables to the stepwise regression models. The performance of the C-3IMERGM estimates was evaluated for different months, elevations, and areas. Compared with the 3IMERGM and 3B43V7 products, the C-3IMERGM dataset exhibited significantly improved performance. The main results are as follows:

1. Both the 3B43V7 and 3IMERGM products overestimated precipitation compared with gauge observations overall, with RB of 10.24% for 3B43V7 and 7.76% for 3IMERGM. 3IMERGM correlated with the gauges better than 3B43V7, with a CC of 0.68 for 3IMERGM and 0.62 for 3B43V7.
2. The C-3IMERGM dataset performed much better than 3B43V7 and 3IMERGM, with more satisfactory metrics. The CC increased from 0.68 for 3IMERGM to 0.73 for C-3IMERGM, and RB decreased from 7.76% for 3IMERGM to − 1.65% for C-3IMERGM. On a monthly scale, the RMSE for C-3IMERGM decreased by 0.11–2.61 mm.
3. For the elevation assessment, the best performance was found in lowland areas (under 1000 m). The 1000–1500 m zone saw the greatest increment for CC, but overestimation of precipitation, whereas precipitation was underestimated at elevations over 1500 m. For the two regions of Xinjiang, the corrected method performed better in SX than in NX. In particular, the underestimation by C-3IMERGM in NX was greater than that by the original 3IMERGM. One possible reason is that the RB for the original 3IMERGM was only − 4.68% in NX, but much higher (35.93%) in SX. However, its CC and RMSE were both obviously improved.
4. Overall, 3B43V7, 3IMERGM, and C-3IMERGM showed similar spatial patterns over Xinjiang, with a declining trend from northwest to southeast. However, considering the sparsity of existing weather stations, C-3IMERGM is more in line with rain gauge data, which is closer to the distribution characteristics of precipitation in Xinjiang.

This study has demonstrated that using correction factors (location and terrain variables) is a practical way to improve the accuracy of the 3IMERGM product in arid regions using a very sparse gauge network. Although the study was conducted in Xinjiang, the correction method could be useful in other arid and semiarid regions around the world, where meteorological and hydrological researchers could achieve more accurate results with corrected precipitation. Future studies should focus on correction of satellite precipitation products at daily or even subdaily scales, which is much more challenging considering the large precipitation variability at finer temporal resolutions.

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